



CY 502: Photochemistry and pericyclic reactions

Course Code	CY 502
Course Name	Photochemistry and pericyclic reactions
L-T-P-C	3-1-0-4
Prerequisites	Organic Chemistry I
Intended for	M.Sc
Distribution	Core
Approval	10th Senate (Jun 2016)

Course Contents

- **Pericyclic Reactions** Molecular orbitals and their symmetry Molecular orbitals and their symmetry properties, Frontier orbitals of ethylene, 1,3-butadiene, 1,3,5- hexatriene and allyl systems. [5 Lectures].
- **Woodward-Hoffmann**
Frontier Molecular Orbital method and Orbital symmetry correlation method. [5 Lectures].
- **Various type of pericyclic reactions**
Electrocyclic reaction; conrotatory and disrotatory motions $4n$, $4n+2$ and allyl systems. Cycloaddition; antarafacial and suprafacial addition, $4n$ and $4n+2$ systems, 2+2 addition of ketenes, Diels-Alder reaction, stereochemical aspects and synthetic applications. 1,3 dipolar cycloadditions and cheletropic reactions. Sigmatropic Rearrangements; suprafacial and antarafacial shifts of H, sigmatropic shifts involving carbon moieties, 3,3- and 5,5- sigmatropic rearrangements, Claisen, Cope and Aza-Cope rearrangements. Ene reaction. [15 Lectures].
- **Photochemistry**
Quantum yields, intersystem crossing, photosensitization and energy transfer reactions. Photochemistry of olefins and carbonyl compounds, photo oxygenation and photo fragmentation, Photochemistry of aromatic compounds: isomerisation, additions and substitutions. Singlet molecular oxygen reactions. Paterno-Buchi reaction, Di-pi-methane rearrangement, Bartons reaction and Photo-Fries rearrangement. [17 Lectures].

Text Books

1. Charles H. DePuy, **Molecular Reactions and Photochemistry**, Orville Lamar Chapman
2. I. Fleming, **Frontier Orbital and Organic Chemical Reactions**, John Wiley, 1976.
3. W. Carruthers, **Some modern Methods of Organic Synthesis**, Cambridge University Press, 1990.
4. T.W. Greene, **Protective Groups in Organic Synthesis**, Wiley-VCH, 1999.

References

1. T. L. Gilchrist and R. C. Storr, **Organic Reactions and Orbital Symmetry**, 2nd Edition, CUP.
2. L. A. Paquette, **Modern Heterocyclic Chemistry**, W.A. Benjamin, Inc., 1968.
3. Ian Fleming, **Pericyclic reactions**, Oxford University Press, 1999.
4. S. Sankararaman, **Pericyclic Reactions: A Textbook**, Wiley-VCH, 2005
5. N. J. Turro, V. Ramamurthy, J. C. Scaiano, **Modern Molecular photochemistry of organic molecules**, University Science books, 2010